

**2017 Aptitude 2nd Round
Entrance Exam Answer**

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EDUCATIONAL ASSESSMENT AND EXAMINATIONS SERVICE (EAES) ETHIOPIAN SECONDARY
SCHOOL LEAVING CERTIFICATE EXAMINATION (ESSLCE) Scholastic Aptitude Test, Sene, 2017
E.C / June / July, 2025 G.C

NUMBER OF ITEMS: 60

TIME ALLOWED: 2 HOURS

SUBJECT CODE: 03

BOOKLET CODE: 369



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Section One: Verbal Reasoning (Questions 1 - 35)

Directions: This section, which is subdivided into Analogy, Synonyms, Antonyms, Classification, Sentence, Completion, Reading Comprehension, and Analytical and Logical Reasoning, contains 35 items altogether. Each subsection is followed by its specific instruction.

Instruction: For items 1-4, read each of the Pair of words written in capital letters carefully and choose the one which has a similar relationship with the given word. Then, blacken the letter of your choice on the separate answer sheet provided.

1. EDUCATION: INVESTMENT; Security:

- A. Warfare
- B. Peace
- C. Power
- D. Jail

2. FOOD: LIVELIHOOD; Accountability:

- A. Dictatorship
- B. Friendship
- C. Relationship
- D. Leadership

3. INDIVIDUAL: CITIZENSHIP; State:

- A. Authorship
- B. Governance
- C. Democracy
- D. Aristocracy

4. RIGHT: MISTAKEN; Life:

- A. Contentment
- B. Gratification
- C. Miserable
- D. Progress

Instruction: Questions 5 - 8 consist of words written in capital letters followed by four alternatives. Choose the word that is most nearly similar in meaning to the word in capital. Then, blacken the letter of your choice on the separate answer sheet provided.

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5. RESILIENCE

- A. Fragility
- B. Elasticity
- C. Depression
- D. Vibrancy

6. PROXIMITY

- A. Neighborhood
- B. Distance
- C. Demarcation
- D. Polarity

7. RETALIATION

- A. Conspiracy
- B. Amnesty
- C. Revenge
- D. Compassion

8. ABUNDANCE

- A. Prosperity
- B. Destitution
- C. Scarcity
- D. Poverty

Instruction: Questions 9 - 12 consist of words written in capital letters followed by four alternatives. Choose the word that is most nearly opposite in meaning to the word in capital. Then, blacken the letter of your choice on the separate answer sheet provided.

9. FORWARD

- A. Onward
- B. Backward
- C. Progress
- D. Ideal

10. SELECT

- A. Discard

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B. Upgrade

C. Pick

D. Undermine

11. LOOSE

A. Give up

B. Withdraw

C. Aspire

D. Tight

12. ELECT

A. Choose

B. Dismiss

C. Abstain

D. Discriminate

Instruction: Questions 13 - 16 contain category of words one of which does not belong to the group. Choose a word that does NOT fit into the group and blacken the letter of your choice on the separate answer sheet provided.

13. A. Crime

B. Court

C. Law

D. Jury

14. A. Principal

B. School

C. Pupil

D. Learner

15. A. Spouse

B. Husband

C. Wife

D. Sibling

16. A. Pilot

B. Sailor

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C. Passenger

D. Hostess

Instruction: Each sentence below 17 - 20 has one blank space followed by four options. Choose the option that best completes the contextual meaning of the sentence and blacken the letter of your choice on the separate answer sheet provided.

17. A sustainable reconciliation between the warring parties would hopefully contribute to the _____ of peace and social security across the country.

A. disruption

B. destabilization

C. restoration

D. violation

18. The _____ of Africans into Western Christianity was the byproduct of Western colonialism.

A. exploitation

B. trafficking

C. dehumanization

D. conversion

19. Abraham Lincoln was the 16th American President who championed the _____ of the slave trade during the Civil war against the separatist southern states.

A. abolition

B. endorsement

C. adoption

D. expansion

20. The _____ of ICT in our contemporary world has radically transformed the modes of interpersonal and international communication.

A. coincidence

B. emergence

C. translation

D. crisis

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Instruction: Read the following Passage carefully and answer questions 21 - 24 on the basis of the information stated or implied in the passage. Then, blacken the letter of your choice on the separate answer sheet provided.

Passage I

Intellectuals, who are visionary citizens endowed with profound wisdom and foresight, apply their knowledge for the purpose of awakening society with a view derived from the Vietnamese word for intellectual, trithuc, which denotes the blend of "mind" and "awaken".

There are three key factors in nurturing intellectuals: knowledge; ability and willingness to awaken society to a noble cause. It is evident that discussing the role of intellectuals is as good as discussing the social responsibility of intellectuals. Einstein maintains that "The world is a dangerous place not because of those who do evil, but because of those who look on and do nothing." Intellectuals should thus mobilize their capacity to steering society away from the foreseeable negative impacts not only just for past multifarious efforts, but also for current and future inappropriate endeavours, and protecting social standards and universal values.

(Source: Slightly adapted from Gian Tu Trung: What is the Role of Intellectuals in Society? World Economic Forum 15 March 2013)

21. According to the passage, who is an intellectual?

- A. A person with a terminal degree
- B. Somebody with bookish learning
- C. Someone who discharges social responsibility
- D. A person neutral to the problems of a country

22. Which one of the following attributes characterizes intellectuals?

- A. Pragmatism and Vision
- B. Providentialism and Opportunism
- C. Neutrality and Subjectivity
- D. Isolation and Profound wisdom

23. What does Einstein's critical observation imply?

- A. The responsibility of policy makers to society
- B. The subjective role of pseudo intellectuals.
- C. The negative impacts of psycho passive individuals

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- D. The resemblance between evil doers and intellectuals
24. The scope of intellectual intervention primarily focuses on
- A. creating awareness in a society.
 - B. westernization of values
 - C. drumming about domestic issues.
 - D. upholding traditional views.

Reading Comprehension

Instruction: Read the following passage carefully and answer questions 25 - 28 on the basis of the information stated or implied in the passage. Then, blacken the letter of your choice on the separate answer sheet provided.

Passage II

Fiscal Policy and African Development

(1.) Fiscal policy deals with the use of financial resources; and has the potential to be a key driver of Africa's growth and development (2). However, the fiscal reforms and adjustments introduced since 2000 had a mixed impact on fiscal performance across countries. (3) In many other countries, increased government revenue supported investment in infrastructure and services, fostering economic growth. (4) However, several African countries are experiencing persistent fiscal deficits and a narrowing fiscal space with high and rising debt, threatening macroeconomic stability.

(5.) Most countries continue to practice a cyclical fiscal policy. (6) To reduce fiscal vulnerability, countries need to change their fiscal practices. (7) Countries could increase their tax revenue by 4 to 7 per cent of GDP by shifting from a cyclical to counter cyclical fiscal policy. (8) African governments should therefore put more effort into strengthening macroeconomic management and improving spending efficiencies. (9) African countries are advised to reallocate more funds to health and education to achieve higher investment and growth.

(10) Fiscal policy can also crowd in private investment in infrastructure and help encouraging research and development enhancing the business environment, (11) African governments should also take part in the ongoing global race to the bottom, as countries rush to cut corporate taxes to enhance competitiveness and attract investment. (12) Taxes on income, profits and capital gains could have to be cut by half to increase total investment by just 1 per

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cent. (13) Fiscal policy can second largest impact on increasing the value of manufacturing (14) The factors affecting fiscal performance are interconnected and thus require a holistic policy framework to address them.

Source: United Nations Economic Commission for Africa Fiscal Policy for Financing Sustainable Development in Africa.

25. According to the passage, many African countries are experiencing _____

- A. Growing debt and financial scarcity.
- B. fast growth of infrastructure.
- C. increased government revenue.
- D. high level progress of macro economy.

26. Which sentence in the text is used as evidence for the answer to question number 25?

- A. Sentence 12 [Taxes on income ... by just one percent.]
- B. Sentence 11 [African governments ... attract investment.]
- C. Sentence 5 [Most countries ... practice cyclical fiscal policy.]
- D. Sentence 4 [...African countries...macroeconomic stability.]

27. What should African countries do to have wise use of their finance?

- A. They should enhance taxation by lessening tax rate on profits.
- B. They should allocate more finances on health and education sectors.
- C. They have to manage and spend their economic resources properly.
- D. They should promote consumption of goods and service in all sectors.

28. Which sentence in the reading text is used as evidence for the answer to question number 27?

- A. Sentence 10 [Fiscal policy... business environment.]
- B. Sentence 8 [African governments... efficiencies.]
- C. Sentence 12 [Taxes on income... just one percent.]
- D. Sentence 13 [Fiscal policy... manufacturing.]

Analytical and Logical Reasoning

Instruction: Questions 29 - 35 deal with analytical and logical reasoning. Read the given set of conditions and assumptions for each question. Then, choose the answer that you think is most

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appropriate among the given alternatives and blacken the letter of your choice on the separate answer sheet provided.

29. **Statement I:** Some countries do not incorporate Civic and Moral Education in the national curriculum of their countries.

Statement II: Juvenile delinquency and drug addiction is growing by the day in such countries.

What is the relationship between the above statements?

- A. Statement I refers to the consequences of neglecting ethical and moral education.
- B. Statement II asserts the root causes of juvenile delinquency and drug addiction.
- C. Statement I refers to the cause while statement II relates to the effect.
- D. There is no logical relationship between the two statements.

Section Two: Quantitative Reasoning (Questions 36 - 60)

Directions: This section contains 25 quantitative reasoning items which consists of algebra, sequence, geometry, logical analytics reasoning, probability and statistics. Solve each of the problems given in questions 36 - 60 according to their specific instructions and select the correct answer from the given four alternatives. Then, blacken the letter of your choice on the separate answer sheet provided.

36. Which one of the following is equal to $-6 + (1+3^3)/(2 \times 7)$?

- A. $77/4$
- B. 32
- C. 92
- D. 56

37. Which one of the following triple of numbers contains an irrational number?

- A. 3.01, $22/7$, $\sqrt{36}$
- B. 3.14, $\sqrt{121}$, 2.01001
- C. 2.1313, 3.1412, $7/\sqrt{484}$
- D. 3.54, $\sqrt{55}$, $1/3$

38. If a and b are two real numbers such that: $2a^2 - 2b^2 = 8640$ and $a - b = 120$; then what is the value of $a + b$?

- A. 360
- B. 36

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C. 72

D. 240

39. Which one of the following is equal to the ratio 36 km/hr to 30m/s?

A. 1:3

B. 3:1

C. 5:6

D. 6:5

40. Which one of the following pairs of equations are equivalent?

A. $3x + 2 = -1$; $6x + 6 = 0$

B. $x/2 + 4 = 1$; $x + 6 = -1$

C. $x/3 - 1/3 = 0$; $2x + 6 = 0$

D. $3x + 5 = -x + 9$, $4x = -4$

41. A telecom company provides a package of Birr 500 service fee per month plus birr 0.75 per minute air time. If a man used 2000 minutes air time in one month, then how much birr will he pay to the telecom company for that specific month?

A. 1500

B. 2000

C. 2000

D. 1750

42. If x , y and z are real numbers such that $x + y - z = 300$ and $x + y + 2z = 900$; then that is the value of $x + y$?

A. 600

B. 100

C. 100

D. 500

43. If X is a real number and $f(x) = 3x^2 + 5x + \sin x$ then what is the value of $3f(-1) - 2f(0)$?

A. -4

B. -6

C. 3

D. 8

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44. The temperature $T(t)$ (in degree Celsius) of a cup of coffee can be modeled by the equation $T(t) = 87 - 3t^2$ where t is the time in minutes after the coffee was poured into the cup. How long will it take in minute for the coffee to cool down to 60 degree Celsius?

- A. 5
- B. 2
- C. $10/3$
- D. 3

45. What is the value of x in the sequence 25, 4, 49, 16, 81, 36, 121, 64, x , ...?

- A. 169
- B. 144
- C. 225
- D. 100

46. A car driver drove the first 100 km out of 400 km distance with a constant speed of 200 km/hr. Given he changed his speed to twice as fast at the first speed to drive the next 100km, and kept on changing the speed in the same manner in every 100km until he reached his final destination. Which one of the following is true?

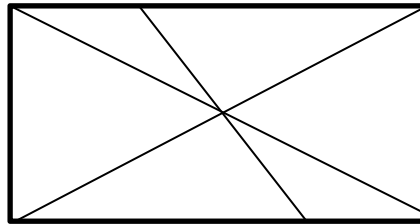
- A. The speed of the car is fastest 100 kms was 103 km/hr.
- B. The car needed 4 hours to cover the given 400 kms distance.
- C. The driver uses less than 2 hours to cover the first 300 kms.
- D. The average speed of the driver for the whole distance was 165 km/hr.

47. Which one of the following is the total surface area (in cm^2) of a regular triangular pyramid of whose lateral surfaces are equilateral triangles of side length 10cm each?

- A. $75\sqrt{3}$
- B. $25\sqrt{3}$
- C. $100\sqrt{3}$
- D. $50\sqrt{3}$

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48. In the following figure how many triangles are there?



- A. 12
 - B. 13
 - C. 8
 - D. 6
49. From a jar containing 5 red balls and 7 blue balls of the same size, a player drew 2 balls at once. What is the number of ways of drawing one blue ball and one red ball?
- A. 12
 - B. 35
 - C. 21
 - D. 66
50. If two six faced dice are rolled, then which one of the following is the probability of getting two numbers whose sum is greater than or equal to 5?
- A. $\frac{5}{6}$
 - B. $\frac{3}{4}$
 - C. $\frac{11}{12}$
 - D. $\frac{7}{9}$
51. Suppose the symbols ♠, ♥, and ♦ represent three one digit numbers and each symbol has the same value all the time. If the product $(2♠♥)1♦$ gives 4674 and the sum $♦4♥+1♦5+2♠$ gives 836, then which of the following are the possible value of ♠, ♥ and ♦ respectively?
- A. 4, 8 and 9
 - B. 6, 4 and 9
 - C. 4, 4 and 6
 - D. 9, 4 and 6

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52. Four persons M, N, T and P work together to complete the same task. If M, N and T worked $(2/7)^{\text{th}}$, $(1/7)^{\text{th}}$ and $(2/9)^{\text{th}}$ of the task respectively, and P completed the remaining part of the task, then what is the part of task that P completed more than T?

- A. $22/63$
- B. $8/63$
- C. $2/3$
- D. $41/63$

Instructions: For Questions 53 and 54 using the information given in the table choose the correct answer from the given alternatives.

<i>Water tanks</i>	<i>Total Carrying capacity in Liters</i>	<i>Number of pipes surroundings the tank</i>	<i>Amount of matter in Liters that passes through each pipe in 20 minutes</i>
<i>T1</i>	<i>75,900</i>	<i>6</i>	<i>275</i>
<i>T2</i>	<i>95,680</i>	<i>8</i>	<i>260</i>

53. Assuming that the tanks were empty and both the tanks started receiving water from all the pipes surrounding them at the same time, which one of the following is NOT true?

- A. Both tanks took equal length of time to be filled.
- B. Tanker T_1 requires 15 hours and 20 minutes to be filled.
- C. T_2 carries 62,400 liters in 10 hours.
- D. Tanker T_2 carries 75,250 liters in 15 hours.

54. If all the pipes surrounding T_1 were working together and two of them stopped working after 5 hours, then how many liters of water is in T_1 after a total of 12 hours?

- A. 24,750
- B. 47,850
- C. 23,100
- D. 59,400

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55. A student increased her usual speed by 50% and reached at her school 20 minutes earlier than her usual time of arrival using the same path. Which one of the following is the time that could take her to reach the school by travelling with her usual speed (in minutes)?

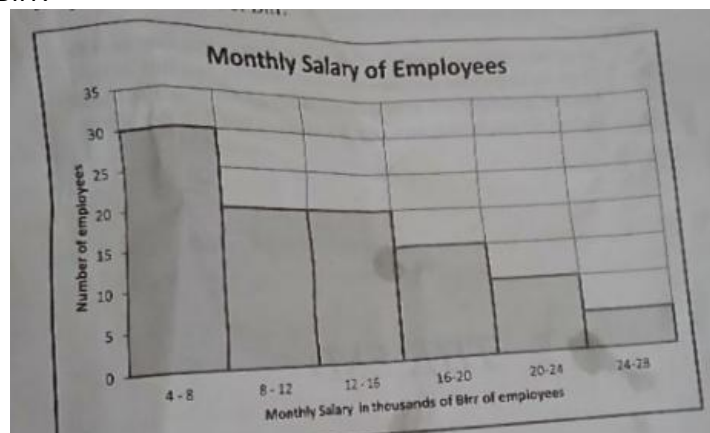
- A. 50
- B. 40
- C. 60
- D. 70

56. In a game of 80 points, where T and R are playing together. T has given 16 points to R. When M and T are playing together. T had given 20 points to M. Which one of the following is the number of points that M has to give to R when they play together in a game of 150 points?

- A. 50
- B. 40
- C. 30
- D. 60

Instructions: For Questions 57 – 60 use the following information and answer each question based on the information therein.

The following histogram presents the monthly salary of employees at a certain company in thousands of Birr.



57. What is the number of employees who earn monthly salary of less than Birr 16,000?

- A. 50
- B. 60
- C. 70

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D. 65

58. Which one of the following is the average monthly salary of all the employees (in Birr)?

A. 12,068

B. 12,800

C. 12,000

D. 12,008

59. What percent of the employees earn monthly salary of Birr 8,000 and above, but less than Birr 20,000?

A. 55%

B. 50%

C. 45%

D. 35%

60. Which one of the following is true about the salary distribution in the given histogram?

A. 15 of the employees earn monthly salary of more than Birr 16,000.

B. 25 of the employees earn monthly salary of more than Birr 20,000.

C. Most of the employees earn monthly salary of less than Birr 16,000.

D. Half of the employees earn monthly salary of less than Birr 8,000.

ANSWER KEY

Section One: Verbal Reasoning (Questions 1-35)

1. **B. Peace**

(Analogy: Security is a state that promotes peace, similar to education being an investment.)

2. **D. Leadership**

(Analogy: Accountability is essential for leadership, similar to food being essential for livelihood.)

3. **B. Governance**

(Analogy: A state is characterized by governance, similar to an individual having citizenship.)

4. **C. Miserable**

(Analogy: Life can be miserable as an opposite state, similar to right being mistaken.)

5. **B. Elasticity**

(Synonym: Resilience means the ability to recover, similar to elasticity.)

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6. **A. Neighborhood**

(Synonym: Proximity means nearness or closeness.)

7. **C. Revenge**

(Synonym: Retaliation means an act of revenge.)

8. **A. Prosperity**

(Synonym: Abundance means plentiful resources, similar to prosperity.)

9. **B. Backward**

(Antonym: Forward is opposite to backward.)

10. **A. Discard**

(Antonym: Select means to choose; discard means to reject.)

11. **D. Tight**

(Antonym: Loose is opposite to tight.)

12. **B. Dismiss**

(Antonym: Elect means to choose; dismiss means to remove.)

13. **A. Crime**

(Classification: Crime is an act, while court, law, and jury are legal system components.)

14. **B. School**

(Classification: School is an institution, while principal, pupil, and learner are people.)

15. **D. Sibling**

(Classification: Spouse, husband, and wife are marital terms; sibling is a blood relation.)

16. **C. Passenger**

(Classification: Pilot, sailor, and hostess are occupations; passenger is a role.)

17. **C. restoration**

(Sentence completion: Reconciliation contributes to restoring peace.)

18. **D. conversion**

(Sentence completion: Colonialism led to conversion to Western Christianity.)

19. **A. abolition**

(Sentence completion: Lincoln championed the abolition of the slave trade.)

20. **B. emergence**

(Sentence completion: The emergence of ICT transformed communication.)

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21. C. Someone who discharges social responsibility

(Reading Comprehension: Intellectuals apply knowledge to awaken society.)

22. A. Pragmatism and Vision

(Reading Comprehension: Intellectuals have vision and wisdom to steer society.)

23. C. The negative impacts of psycho passive individuals

(Reading Comprehension: Einstein implies passive individuals contribute to danger.)

24. A. creating awareness in a society.

(Reading Comprehension: Intellectuals focus on awakening society.)

25. A. Growing debt and financial scarcity.

(Reading Comprehension: Countries face fiscal deficits and narrowing fiscal space.)

26. D. Sentence 4

(Evidence for Q25: "However, several African countries are experiencing persistent fiscal deficits...")

27. C. They have to manage and spend their economic resources properly.

(Reading Comprehension: Governments should strengthen management and efficiency.)

28. B. Sentence 8

(Evidence for Q27: "African governments should therefore put more effort into strengthening macroeconomic management...")

29. C. Statement I refers to the cause while statement II relates to the effect.

(Analytical Reasoning: Lack of education causes delinquency and addiction.)

Section Two: Quantitative Reasoning (Questions 36-60)

36. C. 92

37. D. 3.54, $\sqrt{55}$, $\frac{1}{3}$

38. B. 36

39. A. 1:3

40. A.

41. B. 2000 (or C. 2000)

42. D. 500

43. B. -6

44. D. 3

- 45. **A. 169**
- 46. **C. The driver uses less than 2 hours to cover the first 300 kms.**
- 47. **C. $100\sqrt{3}$**
- 48. **A. 12**
- 49. **B. 35**
- 50. **A. $\frac{5}{6}$**
- 51. **C. 4, 4 and 6**
- 52. **B. $\frac{8}{63}$**
- 53. **D. Tanker T2 carries 75,250 liters in 15 hours.**
- 54. **B. 47,850**
- 55. **C. 60**
- 56. **C. 30**
- 57. **C**
- 58. Change to grouped frequency table data the calculate
- 59. _____
- 60. _____

THE END
ENTRANCE TRICKS